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Prevalence of sarcopenic obesity in a Danish population of adults with obesity

Tobias M. Windedal [a,b,c,d], Claus B. Juhl [a,b,c,d], Martin W. Kusk [c,e], Katrine Thelle [c,e], Josje Schoufour [f], Sören Möller [g], Bibi Gram [a,c] Show more

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Highlights

- The prevalence of sarcopenic obesity is low (3%) among adult Danes with obesity.
- No conclusion on sarcopenic obesity and obesity related disease correlations could be drawn.
- The individual clinical impact of sarcopenic obesity in this population remains unclear.

Abstract

Introduction

The prevalence of obesity among adults in Denmark is 18.7%. Sarcopenic obesity, as defined by the combination of low skeletal muscle mass, low muscle strength and obesity, is associated with increased morbidity and early mortality. The prevalence of sarcopenic obesity amongst Danish individuals with obesity is unknown. The South Denmark Obesity Initiative is a screening program where people with a BMI > 30 kg/m^2 and age between 18 and 60 years are examined for the presence of obesity-related diseases. We describe the prevalence of sarcopenic obesity and explore its potential impact in obesity related disease in a Danish cohort of people with obesity.

Methods

Data were collected between September 2020 and April 2025 and included demographic variables (sex, age), anthropometric data (height, weight, hip and waist circumference), assessment of body composition by bioimpedance and single slice CT, hand grip strength, and presence of prediabetes, type 2 diabetes and the metabolic syndrome. The ESPEN/EASO consensus statement criteria for sarcopenic obesity were used.

Results

Of 1053 individuals screened, 1040 were included, of whom 26 females and 5 males met the criteria for sarcopenic obesity.

Conclusion

The prevalence of sarcopenic obesity is low (3.0%) in this Danish screening cohort of people with obesity and with a mean age of 42. The individual impact of having sarcopenic obesity is unclear.

Keywords

Obesity; Sarcopenic obesity; Prevalence; Body composition; Denmark

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1. Introduction

A significant portion of the European population is living with obesity, however, there is variance in rates (average approx. 22%) (Global and regional, 2025; WHO European Regional Obesity Report, 2022) and in Denmark, 18.7% of the adult population are now considered obese with a BMI ≥ 30 kg/m^2 (The health of Danes – The National Health Profile, 2024). The presence of obesity increases the risk of several health threatening conditions. Additionally, individuals with obesity often experience social disadvantages, have more days of sick leave, and report an overall lower health-related quality of life compared to individuals without obesity (Mainous et al., 2016; Luque-Ramirez et al., 2014; Hernaes et al., 2015; George et al., 2018). Reduced physical activity is a well-recognized contributor to obesity, and once obesity develops, further inactivity tends to occur (Martínez-González et al., 1999; Tucker et al., 2013; Silveira et al., 2022). Severe obesity impairs physical function and daily activities, making tasks such as stair climbing, walking, and general mobility increasingly difficult and fatiguing. Physical inactivity diminishes muscle quality, strength, and when combined with obesity, relative muscle strength (strength in proportion to body weight) can be particularly reduced. Obesity-associated impairments in muscle function include decreased oxidative capacity, increased intramuscular lipid accumulation, reduced capillarization, and lower mitochondrial density, collectively contributing to diminished muscular work capacity (Blaak, 2004; de Jongh et al., 2004; Menshikova et al., 2005; Pescatello et al., 2007). Changes in muscle mass and strength substantially impact glucose metabolism and associate strongly with the presence of type 2 diabetes (Han et al., 2025; Xu et al., 2023). Low muscle mass and strength in combination with loss of physical function is defined as sarcopenia, and the coexistence of sarcopenia and obesity is termed sarcopenic obesity (SO) (Donini et al., 2022). The presence of SO is associated with metabolic syndrome, cardiovascular disease, severity of metabolic dysfunction-associated steatotic liver disease, frailty, as well as morbidity and early mortality (Donini et al., 2022; Baygi et al., 2024; Li et al., 2018; Zembura and Matusik, 2022). These associations highlight the key importance of muscle mass and function in obesity.

Insight into SO is relevant as the condition is associated with many diseases and an increased risk of hospitalization and mortality, but population-based evidence on SO in Danish adults has not yet been established. Identification of SO may be used to form strategic initiatives aimed at reducing the risk of developing obesity-, and SO-related diseases. Although weight loss is the cornerstone of obesity management, physical exercise irrespective of weight loss is associated with improvements in, or prevention of, many obesity-related diseases (Martinez-Gomez et al., 2024; Shaw et al., 2006; Mensinger et al., 2016; Elagizi et al., 2020). Subsequently, significant health improvements may be achieved by targeting sarcopenia or low relative muscle strength, rather than exclusively focusing on weight loss. This could add nuance and a more personalized approach to treating complications of obesity.

The South Denmark Obesity Initiative (SDOI) is an ongoing screening program in the Region of Southern Denmark, where individuals with a BMI of more than 30 kg/m^2 and aged between 18 and 60 years are referred by their general practitioner or a hospital department. The program aims to uncover the prevalence of obesity-related diseases, and it reveals undiagnosed obesity-related disease in a large proportion of the participants (Juhl et al., 2024).

2. Aim

Our aim was to describe the prevalence of SO in a Danish population of people with obesity. Furthermore, we aimed to explore and compare the prevalence of prediabetes, type 2 diabetes or metabolic syndrome between the participants with or without SO. Our hypothesis was that the presence of SO correlates with the prevalence of prediabetes, type 2 diabetes or metabolic syndrome.

3. Methods

All the participants in the SDOI screening program are asked if the data collected during the screening may be included in a research database. The screening program in its entirety is described elsewhere (Juhl et al., 2024). In this study, we included demographic variables (age, sex), anthropometric data (height, weight, hip and waist circumference), assessment of body composition by bioimpedance and single slice CT, hand grip strength, as well as screening results regarding the presence of prediabetes, type 2 diabetes and metabolic syndrome.

Data were collected at the University Hospital of Southern Denmark, Esbjerg, between September 2020 and April 2025 after approval by the Region of Southern Denmark (Journal nr. 20-28939 and 22-58053) and the study is registered in the clinicaltrials.gov (NCT05176132) registry. Participation is voluntary and based on informed consent. The study procedures comply with the Declaration of Helsinki.

3.1. Criteria for sarcopenic obesity

Body composition was assessed by bioimpedance analysis, and single slice CT-scan at the level of L3. The muscle function of the participants was assessed by hand grip (HGS) strength measurement. We used the criteria's suggested in the consensus statement, by the European Society for Clinical Nutrition and Metabolism (ESPEN) and the European Association for the Study of Obesity (EASO) to determine the presence of SO assessed by hand-grip strength and bioimpedance analysis (Donini et al., 2022). The ESPEN and EASO consensus statement suggest a formal stepwise screening prior to examination and diagnostic tests for SO. This was not performed prospectively, and the criteria to determine the presence of SO were applied to all the participants included in the study. The criterion for SO is met when a person simultaneously meets the criteria for muscle mass assessment, fat percentage and HGS. The values of cut-off criteria are described under each respective measurement.

Additional assessments were made with single slice CT-scan.

Furthermore, the ESPEN and EASO Consensus Statement stratify SO into two stages. Stage I is SO without complication, and stage II is SO with at least one complication attributable to altered body composition and skeletal muscle function (Donini et al., 2022). The complications used to define stage II SO in the SDOI cohort were the presence of either metabolic syndrome, type 2 diabetes, sleep apnoea or liver steatosis. Measurements of these obesity related complications were a part of the screening program.

3.2. Bioimpedance

Bioimpedance was measured using the InBody 770 (InBody Co., Ltd., Seoul, Korea). The bioimpedance output for skeletal muscle mass and fat percentage were reported as kilograms and percentage of total body weight.

Skeletal muscle mass (SMM) was adjusted to body weight as SMM/bodyweight (SMM/W) and reported as percentage. The suggested cut-off values for SMM/W are at < 27.6% for females and < 37.0% for males (Donini et al., 2022). Cut-off values for body fat percentage (PBF) are age dependent at ≥39% for females and ≥26% for males, aged 20 to 39 years, ≥41% for females and ≥29% for males aged 40 to 59 years, and ≥43% for females and ≥31% for males aged 60 to 79 years (Donini et al., 2022).

3.3. CT-scans

All CT scans were performed with the SOMATOM® Force (Siemens, Healthineers, Forchheim, Germany). A single, low-dose 5 mm slice was acquired at the level of L3, in inspiratory breath hold. All scans were performed with a fixed tube potential of 120 kVp, to ensure consistent tissue attenuation (Hounsfield Units), with a reference tube current of 160 mAs, automatically adjusted to patient size with automated tube current modulation. Images were reconstructed using a medium smooth convolution kernel (Br32) at a standard extended field of view (FOV) of 78 cm to ensure full axial patient coverage. A trained radiographer performed segmentation of muscle tissue using MIM software© (MIM Software Inc., USA) (version 6.9.7), and all images were reviewed by a radiologist.

Lumbar Skeletal Muscle Index at L3 (LSMIL3) was calculated as muscle area divided by height squared (muscle area/height^2 (cm^2/m^2)). Although the ESPEN and EASO Consensus Statement recommend the use of CT for assessment of body composition when available, no cut-off value is suggested. Since previous studies suggest that LSMI values ≤ 38.5 cm^2/m^2 for women and ≤ 52.4 cm^2/m^2 for men, indicate a relatively low muscle mass (Yamashita et al., 2019; Su et al., 2019; Prado et al., 2008), these are used in present paper.

Furthermore, the measured muscle is reported (cm^2), as well as the visceral adipose tissue (cm^2) to subcutaneous adipose tissue (cm^2) ratio (VAT/SAT ratio).

3.4. Hand grip strength

We used a hand grip dynamometer (Kern MAP 80K1 & MAP 130K1, Kern & Sohn GmbH, Balingen-Frommern, Germany) for HGS measurements. Grip strength was measured in a standing position and with the dynamometer handle in a horizontal position. The average of two attempts, separately for the right and left hand, was noted. Grip strength was considered impaired when the strength was < 16 kg for females and < 27 kg for males (Donini et al., 2022).

4. Criteria for prediabetes, type 2 diabetes and metabolic syndrome

Prediabetes is defined as a HbA1c between 42 and 47 mmol/mol and a plasma glucose ≤ 11.1 mmol/l at the 2-h mark during an Oral Glucose Tolerance Test (OGTT). Type 2 diabetes is determined by either two repeated measurements of HbA1c ≥ 48 mmol/mol, or one measurement above ≥ 42 mmol/mol and a plasma glucose ≥ 11.1 mmol/l at the 2-h mark during an OGTT. Metabolic syndrome is determined by the presence of central obesity (BMI > 30 kg/m^2 or measured waist circumference) and any of two of the following factors: hypertension, high blood glucose levels, high serum triglycerides and reduced serum high-density lipoprotein (HDL). Central obesity is characterized as a waist circumference >88 cm for women and >102 cm for men. Hypertension is defined as a systolic blood pressure above 130 mmHg or a diastolic blood pressure above 85 mmHg. High blood glucose was defined as fasting plasma glucose ≥ 5.6 mmol/L (prediabetes) or ≥ 7.0 mmol/L (diabetes), high serum triglycerides as ≥ 1.7 mmol/L, and reduced HDL cholesterol as < 1.0 mmol/L in men or < 1.3 mmol/L in women (Alberti et al., 2006).

5. Data analysis

A dataset containing demographic characteristics (age, sex) anthropometric data (height, weight, hip and waist circumference), results from assessment of body composition by bioimpedance and single slice CT, hand grip strength, and screening results regarding the presence of prediabetes, diabetes and metabolic syndrome was generated in April 2025. Data were analysed using Stata 18 (Stata©, StataCorp LLC, Texas, US).

The prevalence of SO, prediabetes, diabetes and metabolic syndrome are presented, either as the number or as percentage of participants meeting the criteria or having the condition. Significance level of $\alpha = 0.05$ was set to determine differences between groups.

Mean and standard deviation were calculated for anthropometric data, body composition (bioimpedance and CT) and measurements of hand grip strength. Skewness and kurtosis tests for normality were performed using the Stata package "sktest". For non-normal distributed data, the Wilcoxon rank-sum test was used, while normally distributed data were tested using a two-sample *t*-test with unequal variance.

6. Results

A total of 1053 individuals were screened. Descriptive characteristics of the screened population is given in Table 1.

Table 1. Descriptive characteristics of the study population.

	Study population		
	Female	**Male**	**Total**
N (%)	757 (72)	296 (28)	1053
Age (years)	42.0 (11.1)	43.2 (11.1)	42.3 (11.2)
Height (cm)	167.3 (6.5)	181.6 (7.0)	171.3 (9.2)
Weight (kg)	115.9 (21.1)	136.2 (25.9)	121.6 (24.3)
BMI (kg/m^2)	41.4 (7.1)	41.2 (7.1)	41.3 (7.1)
Waist circumference (cm)	120.1 (15.1)	133.2 (15.3)	123.8 (16.3)
Hip circumference (cm)	133.5 (14.6)	126.4 (13.8)	131.5 (14.7)

Data are reported as numbers (%) and mean (SD).

SD: Standard deviation, N: numbers, BMI: Body mass index.

From the screened population a total of 1040 individuals were included in the statistical analyses.

7. Missing data

In total 13 participants were not included in the statistical analyses due to missing data necessary for the assessment of SO (Bioimpedance, CT and/or HGS).

Nine of the participants did not perform the bioimpedance measurements. For six of the participants, it was due to a pacemaker or an ICD, one participant opted out, and two participants could not complete the measurement due to physical impairments.

Three participants did not receive a CT-scan, two participants opted out, and one participant was not scanned due to a technical error. The results from one participant were not included in the analysis, due to image quality issues.

Three participants did not perform the HGS measurements, two opted out, while one could not perform the test properly due to neuropathy and a recent carpal tunnel surgery.

Data regarding previously known prediabetes, diabetes or metabolic disease were missing for thirteen of the females and five of the male participants. Data from the screening for prediabetes, diabetes and metabolic disease were not available for nine females and seven males.

8. The prevalence of SO

Amongst the participants included in the analyses, 26 females and 5 males met the ESPEN and EASO consensus criteria for SO. This is equivalent to 3.5% of the females (95% CI 2.4 - 5.1%), 1.7% of the males (95% CI 0.7-3.9%) and 3.0% (95% CI 2.1-4.2%) for the total study population.

For the three components of the SO criteria, 720 females (96.4%) and 275 males (93.9%) had a body fat percentage above the cut-off, 380 females (50.9%) and 221 males (75.4%) had a SMM/W below the cut-off, and 47 females (6.3%) and five males (1.7%) had HGS below the cut-off (Fig. 1).

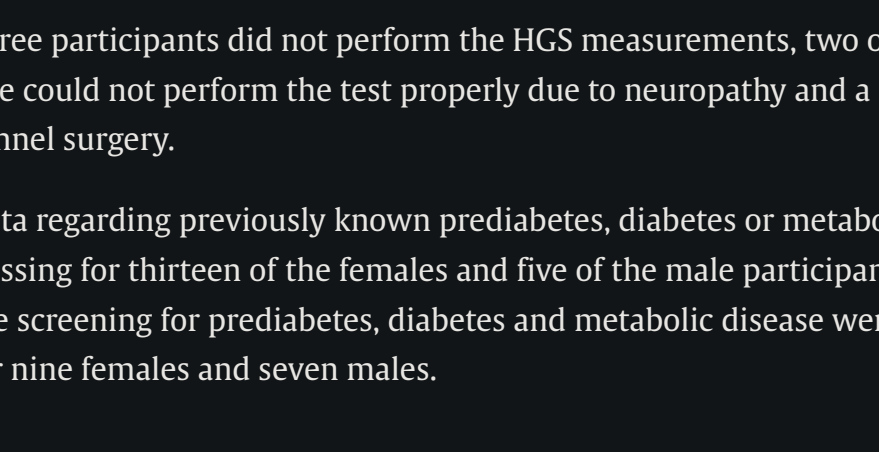

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Fig. 1. Venn-diagram displaying the percentage of included participants from the SDOI study population that met the components of SO.

Fat% indicates the percentage of participants with a body fat percentage above the cut-off. SMM/W indicates the percentage of participants with a bodyweight adjusted skeletal muscle mass below the cut-off. HGS indicates the percentage of participants with a hand grip strength below the cut-off. The overlapping areas indicate the number of participants fulfilling two, or all three SO criteria respectively.

When comparing the participants in the study population who met the SO criteria with the ones that did not, the females with SO were statistically older ($p = 0.05$, 95% CI [−0.0346, −0.00337], Rank-biserial $r = -0.073$), had a significantly higher fat percentage ($p \leq 0.001$, 95% CI [0.0179, 0.0485], Rank-biserial $r = -0.1555$), and a significantly lower muscle mass when measured as either SMM/W ($p \leq 0.001$, 95% CI [−0.056, −0.023], Rank-biserial $r = 0.1868$), muscle area ($p \leq 0.01$, 95% CI [−0.0435, −0.0093], Rank-biserial $r = 0.1242$) or LSMI ($p \leq 0.05$, 95% CI [−0.038, −0.006], Rank-biserial $r = 0.1032$).

The males with SO had significantly lower muscle mass when measured as SMM/W ($p \leq 0.05$, 95% CI [0.083, 1.854] Hedges's $g = 0.972$). There were also indications of a difference in fat percentage with a statistically significant rank-sum exact test ($p \leq 0.05$), however, the 95% CI [−0.003, 0.039] and the effect size indicates a negligible effect (Rank-biserial $r = -0.1196$).

Both the females and the males with SO had significantly lower hand grip strength (Females; right $p \leq 0.001$, 95% CI [−0.0897, −0.042], Rank-biserial $r = 0.308$, left $p \leq 0.001$, 95% CI [−0.009, −0.0363], Rank-biserial $r = 0.275$, Males; right $p \leq 0.001$, 95% CI [1.176, 2.972], Hedges's $g = 2.08$, left $p \leq 0.001$, 95% CI [−0.062, −0.005], Hedges's $g = 0.22$) regardless of dominant hand (Table 2).

Table 2. Comparison of descriptive characteristics and SO assessments of the participants with or without SO in the study population.

	Study population			
	Females		**Males**	
	Non-sarcopenic	**SO**	**Non-sarcopenic**	**SO**
N	721	26	288	5
Age (years)	41.8 (11.0)	45.8 (13.6) *	43.1 (11.1)	37.1 (12.9
Height (cm)	167.4 (6.5)	165.0 (6.0)	181.5 (7.0)	182.2 (8.4
Weight (kg)	115.8 (21.1)	116.2 (21.1)	135.8 (26.0)	147.5 (28.
BMI (kg/m^2)	41.3 (7.1)	42.5 (6.0)	41.2 (7.2)	44.3 (7.6)
Waist circumference (cm)	119.9 (15.1)	122.8 (16.1)	132.9 (15.3)	142.2 (12.
Hip circumference (cm)	133.3 (14.6)	138.0 (15.3)	126.1 (13.7)	136.6 (16.
Bioimpedance				
Fat% (%)	49.1 (4.6)	52.7 (1.8)**	40.2 (7.1)	46.4 (5.8)
SMM/W (%)	28.3 (2.6)	26.0 (1.0)**	33.9 (4.2)	29.9 (3.1)
CT				
Muscle area (cm^2)	150.3 (23.1)	132.2 (23.0)**	216.9 (30.6)	202.3 (44
VAT/SAT (cm^2/cm^2)	0.33 (0.18)	0.34 (0.19)	0.61 (0.33)	0.52 (0.31
LSMIL3 (cm^2/cm^2)	53.8 (8.6)	48.5 (7.4)**	65.9 (9.3)	61.2 (14.0
HGS				
Right hand (kg)	28.9 (6.9)	12.4 (3.0)**	49.3 (10.5)	27.7 (6.4)
Left hand (kg)	26.8 (6.7)	13.8 (5.4)**	45.9 (9.6)	21.2 (4.6)
Dominant hand: Right/Left (n)	641/70	24/2	250/31	3/2
Ambidextrous (n)	10	0	6	0

The results are reported as numbers and mean (SD). Statistical analysis was performed with Wilcoxon rank-sum test, apart from body height, where a two-sample *t*-test with unequal variance was used. Significance level was at $\alpha = 0.05$. P-values; * = $p < 0.05$ ** = $p < 0.01$.

N = number of participants. Fat% = body fat percentage. SMM/W = Skeletal Muscle Mass adjusted by weight. VAT/SAT= Visceral Adipose Tissue to Subcutaneous Adipose Tissue ratio. LSMIL3 = Lumbar Skeletal Muscle Index at L3.

9. The prevalence of prediabetes, type 2 diabetes and metabolic syndrome

One female participant had a documented history of prediabetes prior to participating in the screening program while 24 females and 8 males were diagnosed with prediabetes during screening. None of these met the criteria for SO.

Prior to screening, 17 females and 15 males had a history of type 2 diabetes, of which two females met the criteria for SO. During screening, 17 females and 12 males were diagnosed with type 2 diabetes, none of which met the criteria for SO.

None of the participants were diagnosed with type 1 diabetes during screening, while 9 females and 2 males had a history of type 1 diabetes. None of the participants with type 1 diabetes met the criteria for SO.

During screening, 76 females and 50 males were classified as having metabolic syndrome, of which two females and one male met the criteria for SO.

Further descriptions of how prediabetes, type 2 diabetes and metabolic syndrome were distributed across the participants who either met, or did not meet, the independent components of the SO criteria, can be found in table 3 in the appendix.

The low prevalence of SO (3.0%) did not allow for exploration of any association between SO and the presence of prediabetes, type 2 diabetes or the metabolic syndrome.

9.1. Stratification of SO

Amongst the participants with SO, twelve of the females and four of the males were stratified as having stage II SO with at least one complication attributable to altered body composition and skeletal muscle function.

10. Post hoc analysis

In order to look further into the HGS data, we performed an exploratory post hoc analysis adjusting HGS to BMI. The 10th percentile was selected as a pragmatic, restrictive threshold for this post hoc analysis. As both the threshold and the resulting prevalence estimate were derived from the same study sample, without external validation or optimization against a clinical outcome, this analysis should be interpreted as exploratory and hypothesis-generating only. Categorizing low HGS as performing below the 10th percentile by sex, the BMI adjusted HGS the cut-off values for females were <0.428 for the right hand and <0.430 for the left hand. For the males the corresponding values were <0.802 for the right hand and <0.765 for the left hand. For further details see table 4 in the appendix. This resulted in 55 females (7.4%, 95% CI: 5.7-9.5%) and 29 males (9.9%, 95% CI: 7.0-13.9%) classified as having SO when adjusting HGS for BMI. This is equivalent to a prevalence of 8.1% (95% CI: 6.6-9.9%).

11. Discussion

The prevalence of SO in the study population is 3.0%. The limited number of SO cases precluded a meaningful assessment of the relationship between SO and the metabolic outcomes; prediabetes, diabetes and the metabolic syndrome.

The presence of SO is previously found to be associated with the presence of obesity-related diseases. The high prevalence of obesity-related diseases in the SDOI cohort, and the alterations in muscle mass and function previously observed in obesity gave rise to our hypothesis that SO would be associated with increased risk of prediabetes, diabetes and metabolic syndrome.

If the hypothesis was true, it provided a rationale for strategic initiatives targeting sarcopenia and low relative muscle strength, thereby offering a valuable complement to current weight-loss-centric approaches. However, neither could be tested in the study population, as the prevalence of SO was lower than expected.

Previous studies describe a large variety in the prevalence of SO depending on type of muscle mass assessment, diagnostic criteria and population (Baygi et al., 2024; Zembura and Matusik, 2022; Gao et al., 2021; Benz et al., 2024). In many of these studies, the study population are community-dwelling older adults with higher prevalence of sarcopenia. In comparison, the SDOI study population is younger (mean age 42.3 ± 11.2 years) which most likely contributes to the lower prevalence of SO.

Although the prevalence of SO was too low to draw conclusions regarding the association with prediabetes, type 2 diabetes or metabolic syndrome, it was noteworthy that several of the female participants that met the SO criteria did not have any of these conditions.

Compared with those without SO, women with SO were modestly older and had a higher body fat percentage, lower muscle mass (by SMM/W, muscle area, and LSMI), and markedly lower grip strength in both hands (all $p \leq 0.05$), with generally small effect sizes except for grip strength. Men with SO showed a similar pattern: a small, borderline elevation in body fat percentage, along with significantly lower SMM/W and markedly lower grip strength in both hands (all $p \leq 0.05$). With grip-strength effect sizes considerably larger than for the other components, consistent with hand grip strength being one of the diagnostic criteria used to define SO in this cohort. The lack of a statistically significant difference in muscle area and LSMI between the males with and without SO, is likely due to the low number of male SO cases combined with a higher standard deviation amongst these individuals. Further studies should examine possible sex differences in muscle area, LMSI, muscle outcomes, and functional impairment associated with SO. Some evidence suggests that women with SO may have more pronounced impairments in mobility and basic functional autonomy than men, which could be further investigated using measures of SMM/W and HGS (Lazzer et al., 2026).

Analysing the SO criteria one by one, close to all the participants had a fat percentage above the cut-off value, about three quarters of the males and about half of the females had muscle mass below the suggested cut-off, while only a few had HGS below the cut-off. One possible explanation is that the ESPEN and EASO HGS cut-off might primarily be targeted older populations, thus leading to an underestimation in this cohort. Another possibility is that, even though obesity is associated with alterations in muscle function, absolute handgrip strength (HGS) may not necessarily be lower in individuals with obesity and many individuals with obesity have comparable, or even higher, HGS relative to counterparts without obesity (González et al., 2025; Tomlinson et al., 2016). Consequently, a skeletal muscle mass near the lower end of the reference range may be inadequate for a person with obesity (Walowski et al., 2020). In this cohort, several of the participants performed close to the HGS cut-off value, highlighting the potential need for a more nuanced or graduated set of cut-off values. This led us to perform the post hoc analysis, adjusting HGS to BMI. Even with a conservative cut-off at the 10th percentile, the prevalence of SO is markedly higher at 8.1%. This demonstrates the limitations of absolute HGS as a measure of general muscle function and the test of HGS may have limited ability to identify individuals with SO. The resulting SO prevalence from the post hoc analysis should not be interpreted as a more accurate or clinically validated estimate than the 3.0% obtained using standard ESPEN/EASO absolute cut-offs. Rather, it illustrates the sensitivity of SO prevalence estimates to how impaired muscle strength is operationalized and highlights the need for more nuance in determining impaired physical and/or muscle function. Measuring HGS is quick and easy to perform, does not require expensive equipment, and correlates well with longevity (Benz et al., 2024; Lopez-Bueno et al., 2022). However, in the present study, absolute HGS appears to act as the main limiting factor for the diagnosis of SO. We would therefore argue that in individuals with obesity, it is more accurate to adjust HGS to weight, muscle mass or BMI. Donini et al. (2022) discuss if HGS should be adjusted to body mass in the ESPEN and EASO consensus statement, however, it is not included. Bentz et al. (Benz et al., 2024) found that a normal HGS, while simultaneously having two components of body composition altered (e.g. body fat percentage and skeletal muscle mass) were associated with an increased mortality. Therefore, other tests such as a "sit-to-stand", or a "walking test", might be better measures of physical or muscle function, especially in a younger population as ours, where physical impairments usually are less of a limitation factor. Our post-hoc analysis was exploratory and as such may be used to give ideas to new focus areas.

11.1. Limitations

The results of our study may be limited by selection bias. Although the screening program is open to any individual in the target age group with a BMI above 30 kg/m^2 it might not be a true representation of the background population. Firstly, participation requires a referral from a GP or hospital department. This somewhat limits the generalizability of the results. Second, females are overrepresented (72%). In addition, it is reasonable to think that individuals participating in the SDOI screening program are more aware of their obesity, demonstrate greater interest in their health, and are more motivated to improve their health compared to others in the eligible target population. If so, the participants might also be slightly fitter and have a normal (or closer to normal) muscle function and/or mass than the average of the population eligible to participate in the screening program. This would fit well with our observations, with only very few participants performing below the HGS cut-off criteria. As such, the findings reflect a voluntary screening cohort and should be interpreted accordingly.

The non-participation rate and reasons for non-participation among patients referred to the study were not systematically recorded as part of this study's data collection, which is recognized as a limitation. The lack of prospective formal screening for SO prior to applying the SO criteria is another limitation and could affect the prevalence found in this cohort. Another limitation is that the CT-derived cut-offs have not been specifically validated in a Danish cohort. Furthermore, the study is limited by only including one measure of muscle function (namely HGS).

11.2. Strengths

The study is strengthened by the sample size and the inclusion of both bioimpedance and CT measurements of muscle mass. Although the present cohort is not a random sample of the population, we would argue that the broad inclusion criteria—BMI above 30 and age between 18 and 60 years—make it reasonable to generalize the findings to the broader Danish population of individuals with obesity. The young age of the study population may contribute to a low prevalence of SO. On the other hand, young age may also be seen as a strength, as incidents cases of sarcopenia can be ascribed to obesity rather than age.

12. Conclusion

In this retrospective descriptive study in a Danish screening cohort of people with obesity and mean age of 42 years, we found that the prevalence of SO was 3%. The prevalence of SO was too low to investigate any association with prediabetes, type 2 diabetes or metabolic syndrome.

CRediT authorship contribution statement

Tobias M. Windedal: Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Claus B. Juhl:** Writing – review & editing, Supervision, Project administration, Formal analysis, Conceptualization. **Martin W. Kusk:** Writing – review & editing, Software, Formal analysis, Data curation. **Katrine Thelle:** Writing – review & editing, Data curation. **Josje Schoufour:** Writing – review & editing. **Sören Möller:** Writing – review & editing. **Bibi Gram:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Ethics declaration

Written informed consent to take part in the study and to publish the article has been obtained from all participants or their legal representatives. The privacy rights of participants have been observed.

This study was performed in compliance with relevant laws, regulatory frameworks and guidelines where the research took place. This study was approved by the Regional Committee on Health Research Ethics for Southern Denmark. (Approval No. protocol id S-20210091)

Informed consent statement

All participants in the SDOI cohort received informed consent and specifically consented to have their data used for research purposes. Non-consenters were excluded from analysis.

Institutional review board statement

The study was conducted in accordance with the Declaration of Helsinki and approved by the Regional Committee on Health Research Ethics for Southern Denmark (protocol id S-20210091). Participants were recruited from the South Danish Obesity Initiative (SDOI) research database with written informed consent; the approval specifically covered the additional inclusion of normal-weight controls (BMI < 30) to enable comparisons across weight categories.

Data availability statement

Data may be made available upon reasonable request by contacting the corresponding author.

Patents and intellectual property

There are no patents to disclose.

Other activities

There are no additional activities to disclose.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix.

Table 3. Distribution of prediabetes, type 2 diabetes and metabolic syndrome amongst participants with or without the independent components of the criteria for sarcopenic obesity

	Body fat percentage below cut-off criteria		SMM/W above c
	Females	**Males**	**Females**
N	27	18	367
Prediabetes (n)	0	0	10 (2.7%) (95% CI
History of type 2 diabetes (n)	2 (7.4%) (95% CI: 2.1-23.4%)	0	10 (2.7%) (95% CI
Diagnosed with type 2 diabetes during screening (n)	0	0	10 (2.7%) (95% CI
Metabolic syndrome (n)	3 (11.1%) (95% CI: 3.9-28.1%)	0	30 (2.7%) (95% CI
	Body fat percentage above cut-off criteria		
	Females	**Males**	
N	720	275	
Prediabetes (n)	22 (3.1%) (95% CI:2.0-4.6%)	8 (2.9%) (95% CI:1.5-5.6%	
History of type 2 diabetes (n)	16 (2.2%) (95% CI:1.4-3.6%)	15 (5.5%) (95% CI:3.3-8.8	
Diagnosed with type 2 diabetes during screening (n)	17 (2.4%) (95% CI:1.5-3.8%)	12 (4.4%) (95% CI:2.5-7.5	
Metabolic syndrome (n)	73 (10.1%) (95% CI:8.1-12.6%)	50 (18.2%) (95% CI:14.1-2	

The results are reported as numbers as well as percentage with confidence interval (CI).

Table 4. Post hoc analyses: Hand Grip Strength adjusted to BMI

	Females		Males	
	Right	**Left**	**Right**	**Left**
HGS/BMI	0.70 (0.22)	0.69 (0.21)	1.24 (0.35)	1.14 (0.32)
Percentiles				
10%	0.428	0.430	0.802	0.765
25%	0.559	0.504	0.986	1.010
50%	0.695	0.678	1.222	1.104
75%	0.841	0.809	1.466	1.342
90%	0.966	0.927	1.677	1.532
IQR	0.282	0.305	0.480	0.332

Results reported as mean (SD), percentiles and interquartile range (IQR).

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